

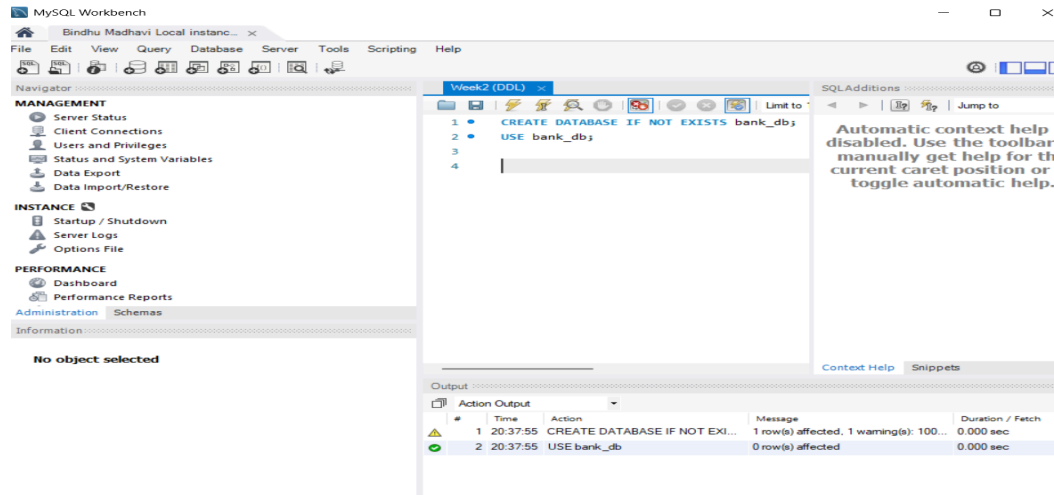
Week-2 (DDL/DCL/TCL)

Name: P. Bindhu Madhavi

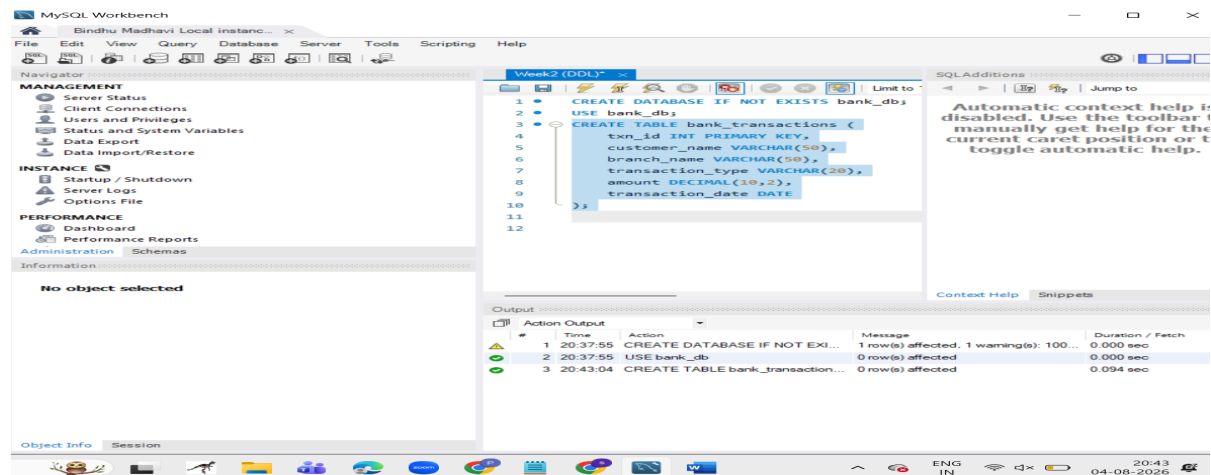
Roll Number: 2520030077

DDL Commands

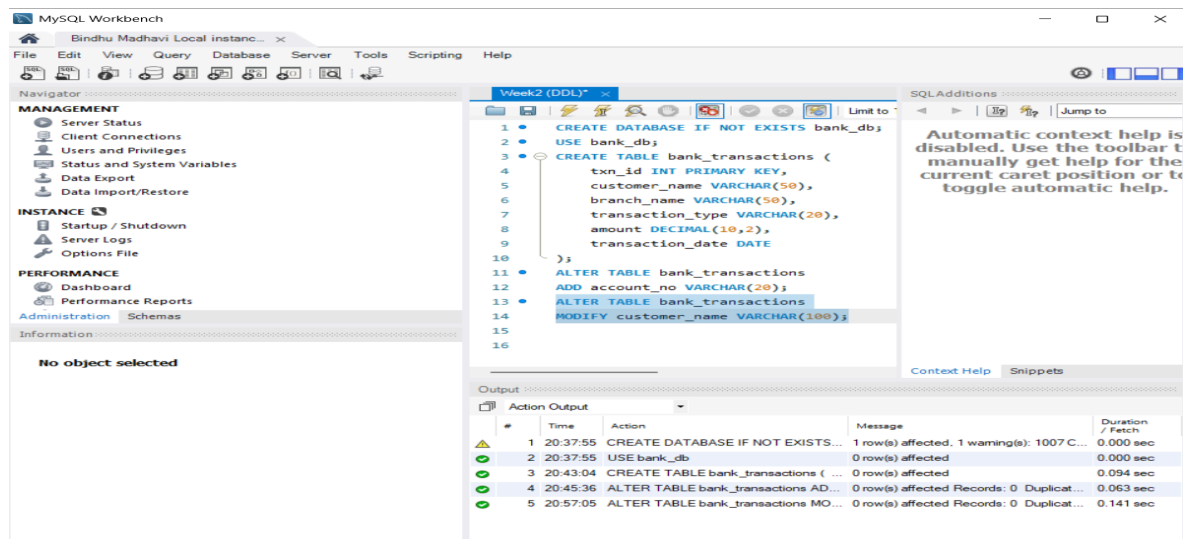
STEP 1: Create Database



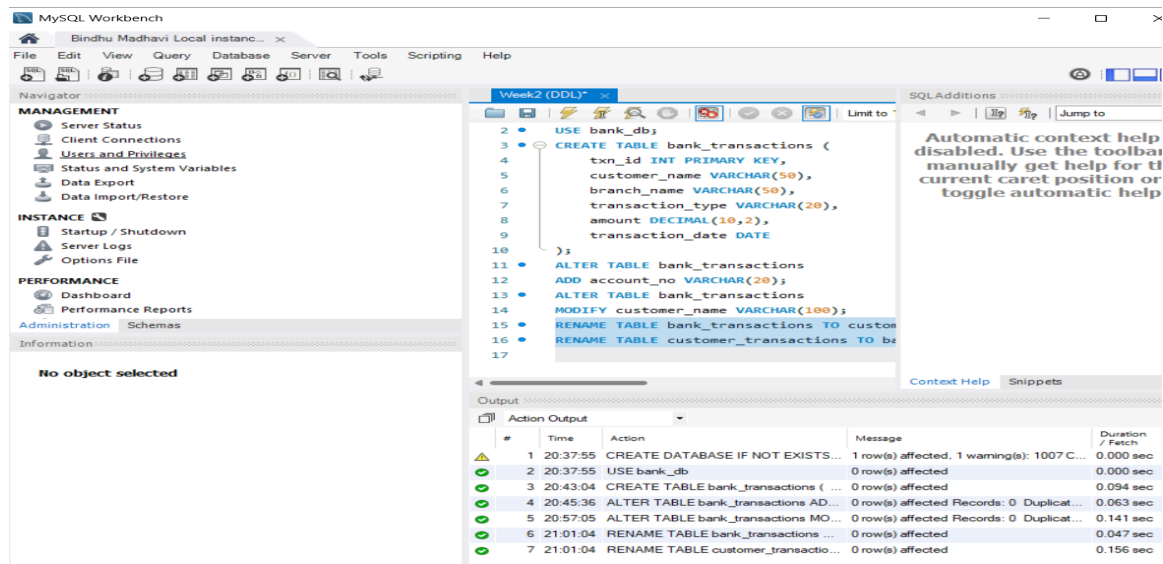
STEP 2: Create Table



STEP 3: Alter Table Add Column



STEP 4: Rename Table



STEP 5: Create Backup Table

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)*

```

2 • USE bank_db;
3 • CREATE TABLE bank_transactions (
4   txn_id INT PRIMARY KEY,
5   customer_name VARCHAR(50),
6   branch_name VARCHAR(50),
7   transaction_type VARCHAR(20),
8   amount DECIMAL(10,2),
9   transaction_date DATE
10 );
11 • ALTER TABLE bank_transactions
12 ADD account_no VARCHAR(20);
13 • ALTER TABLE bank_transactions
14 MODIFY customer_name VARCHAR(100);
15 • RENAME TABLE bank_transactions TO custom
16 RENAME TABLE customer_transactions TO be
17 • CREATE TABLE bank_backup AS
18 SELECT * FROM bank_transactions;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	20:37:55	CREATE DATABASE IF NOT EXISTS...	1 row(s) affected, 1 warning(s): 1007 C...	0.000 sec
2	20:37:55	USE bank_db	0 row(s) affected	0.000 sec
3	20:43:04	CREATE TABLE bank_transactions (...	0 row(s) affected	0.094 sec
4	20:45:36	ALTER TABLE bank_transactions AD...	0 row(s) affected Records: 0 Duplicat...	0.063 sec
5	20:57:05	ALTER TABLE bank_transactions MO...	0 row(s) affected Records: 0 Duplicat...	0.141 sec
6	21:01:04	RENAME TABLE bank_transactions ...	0 row(s) affected	0.047 sec
7	21:01:04	RENAME TABLE customer_transactio...	0 row(s) affected	0.156 sec
8	21:03:46	CREATE TABLE bank_backup AS S...	0 row(s) affected Records: 0 Duplicat...	0.063 sec

Object Info Session

STEP 6: Insert Sample Data

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

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INSTANCE

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PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)*

```

16 • RENAME TABLE customer_transactions TO be
17 • CREATE TABLE bank_backup AS
18 SELECT * FROM bank_transactions;
19 • INSERT INTO bank_transactions
20 (txn_id,customer_name,branch_name,transa
21 VALUES
22 (101,'Ravi','Hyderabad','Deposit',5000,'
23 (102,'Sita','Hyderabad','Withdrawal',200
24 (103,'Kiran','Vijayawada','Deposit',1200
25 (104,'Anil','Vizag','Deposit',8000,'2024
26 (105,'Priya','Hyderabad','Withdrawal',3
27 (106,'Ramesh','Vizag','Deposit',15000,'
28 (107,'Keerthi','Vijayawada','Withdrawal'
29 (108,'Rahul','Hyderabad','Deposit',9000,
30 (109,'Sneha','Vizag','Withdrawal',4000,
31 (110,'Madhu','Vijayawada','Deposit',1100

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

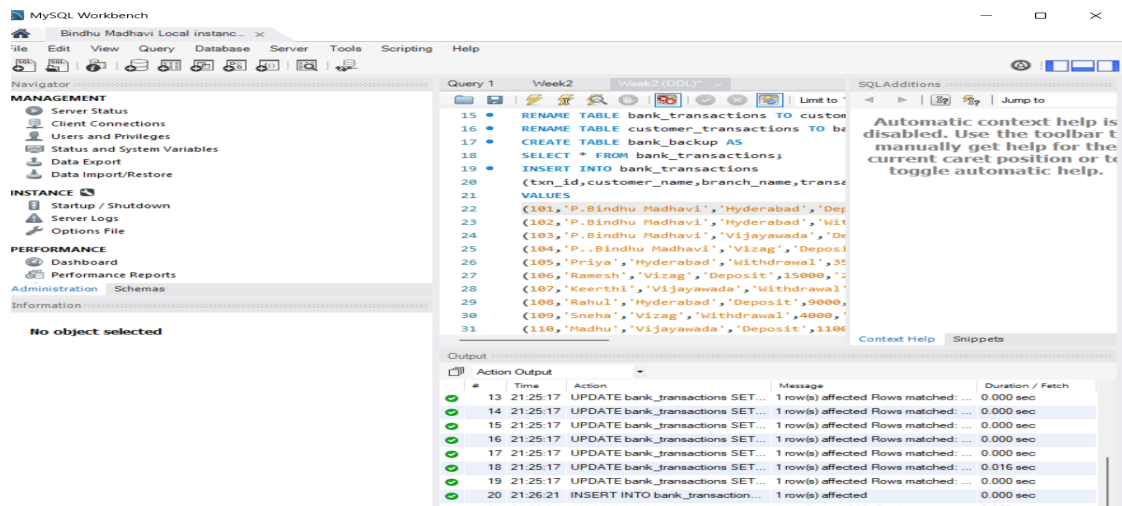
Output

Action Output

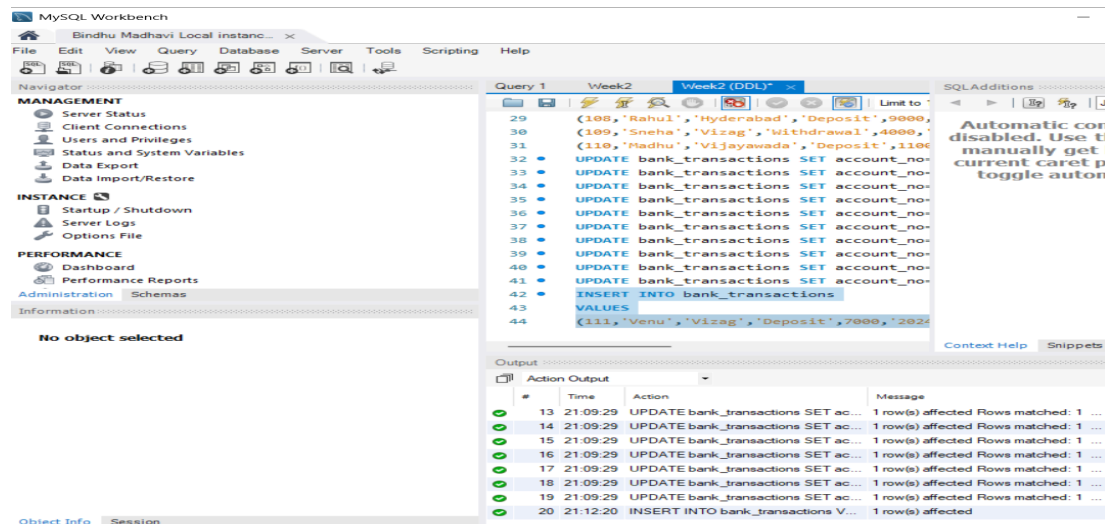
#	Time	Action	Message	Duration / Fetch
2	20:37:55	USE bank_db	0 row(s) affected	0.000 sec
3	20:43:04	CREATE TABLE bank_transactions ...	0 row(s) affected	0.094 sec
4	20:45:36	ALTER TABLE bank_transactions ...	0 row(s) affected Records: 0 Duplicat...	0.063 sec
5	20:57:05	ALTER TABLE bank_transactions ...	0 row(s) affected Records: 0 Duplicat...	0.141 sec
6	21:01:04	RENAME TABLE bank_transactions...	0 row(s) affected	0.047 sec
7	21:01:04	RENAME TABLE customer_transact...	0 row(s) affected	0.156 sec
8	21:03:46	CREATE TABLE bank_backup AS ...	0 row(s) affected Records: 0 Duplicat...	0.063 sec
9	21:06:38	INSERT INTO bank_transactions (x...	10 row(s) affected Records: 10 Dupl...	0.016 sec

Object Info Session

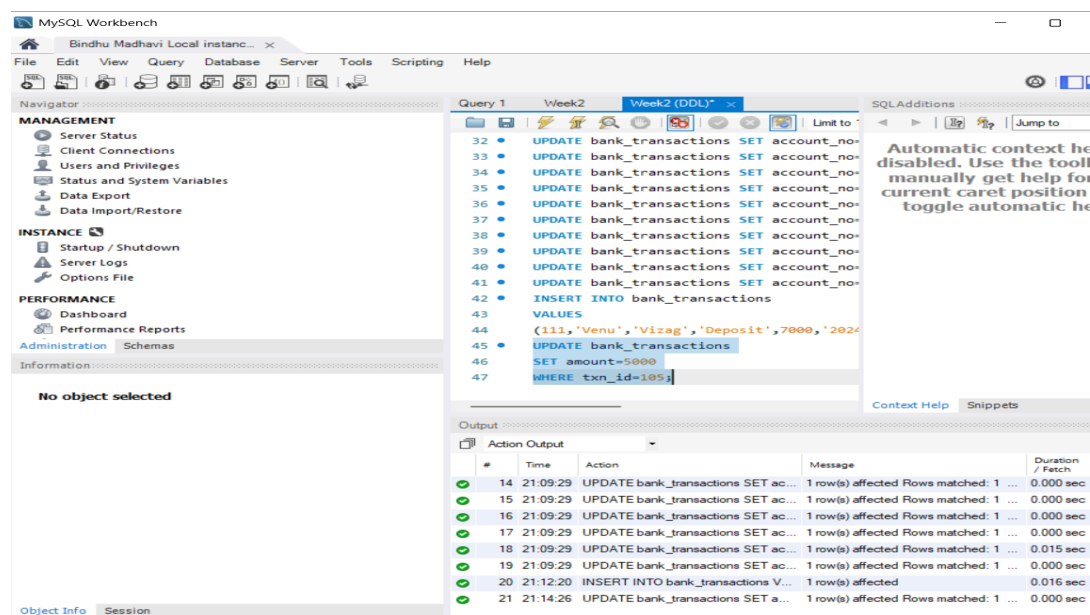
STEP 7: Update Account Numbers



STEP 8: Insert



STEP 9: Update



STEP 10: Delete

The screenshot shows the MySQL Workbench interface. The left sidebar contains the Navigator pane with sections for MANAGEMENT, INSTANCE, and PERFORMANCE. The main editor displays a SQL query in the 'Query 1' tab, which includes several UPDATE statements followed by a DELETE statement: `DELETE FROM bank_transactions WHERE txn_id=111;`. The 'Output' pane at the bottom shows the execution results, including a table with columns for #, Time, Action, Message, and Duration / Fetch. The message for the DELETE statement indicates '1 row(s) affected'.

#	Time	Action	Message	Duration / Fetch
15	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
16	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
17	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
18	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.015 s
19	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
20	21:12:20	INSERT INTO bank_transactions V...	1 row(s) affected	0.016 s
21	21:14:26	UPDATE bank_transactions SET a...	1 row(s) affected Rows matched: 1 ...	0.000 s
22	21:15:59	DELETE FROM bank_transactions ...	1 row(s) affected	0.015 s

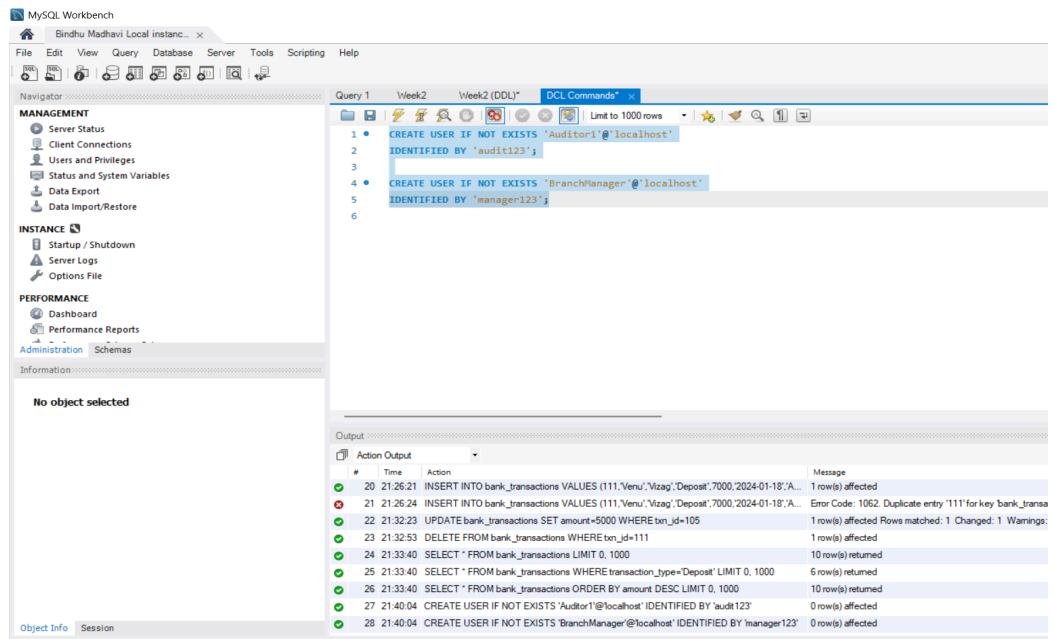
STEP 11: Select

The screenshot shows the MySQL Workbench interface. The main editor displays a SQL query in the 'Query 1' tab, which includes a WHERE clause followed by a SELECT statement: `SELECT * FROM bank_transactions;`. The 'Result Grid' pane shows the execution results, displaying a table with columns: txn_id, customer_name, branch_name, and trans. The 'Output' pane at the bottom shows the execution results, including a table with columns for #, Time, Action, Message, and Duration / Fetch. The message for the SELECT statement indicates '10 row(s) returned'.

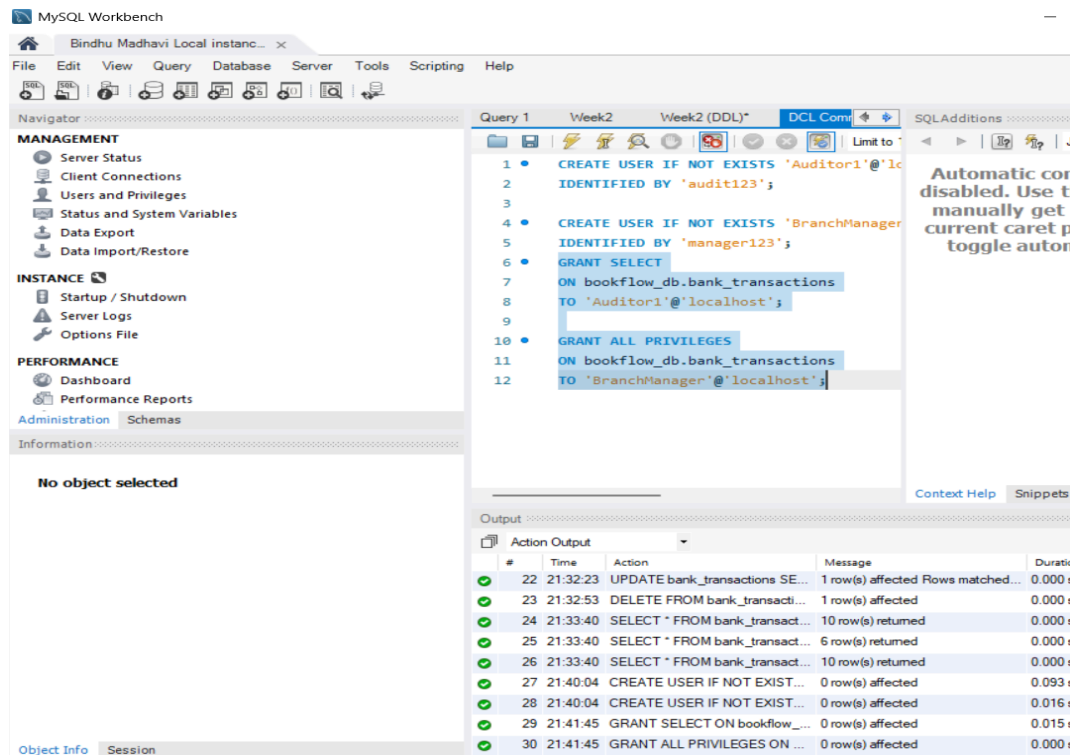
txn_id	customer_name	branch_name	trans
106	Ramesh	Vizag	Depo
103	P.Bindhu Madhavi	Vijayawada	Depo
110	Madhu	Vijayawada	Depo
108	Rahul	Hyderabad	Depo
104	P..Bindhu Madhavi	Vizag	Depo
101	P.Bindhu Madhavi	Hyderabad	Depo
105	Priya	Hyderabad	Withc
109	Krusha	Vizag	Withc

DCL COMMANDS

STEP 1: Create User - Auditor1 && Branch Manager



STEP 2: Grant SELECT Permission



STEP 3: Check Current User

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

MANAGEMENT

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- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Object Info Session

Week2 Week2 (DDL)* DCL Commands*

```

1 CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
2 IDENTIFIED BY 'audit123';
3
4 CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'
5 IDENTIFIED BY 'manager123';
6 GRANT SELECT
7 ON bookflow_db.bank_transactions
8 TO 'BranchManager'@'localhost';

```

Result Grid

Filter Rows:

Result Grid

Form Editor

Result 1 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
23	21:32:53	DELETE FROM bank_transacti...	1 row(s) affected	0.000 sec
24	21:33:40	SELECT * FROM bank_transact...	10 row(s) returned	0.000 sec
25	21:33:40	SELECT * FROM bank_transact...	6 row(s) returned	0.000 sec
26	21:33:40	SELECT * FROM bank_transact...	10 row(s) returned	0.000 sec
27	21:40:04	CREATE USER IF NOT EXIST...	0 row(s) affected	0.093 sec
28	21:40:04	CREATE USER IF NOT EXIST...	0 row(s) affected	0.016 sec
29	21:41:45	GRANT SELECT ON bookflow...	0 row(s) affected	0.015 sec
30	21:41:45	GRANT ALL PRIVILEGES ON ...	0 row(s) affected	0.000 sec
31	21:49:14	SELECT CURRENT_USER() LI...	1 row(s) returned	0.000 sec

Automatic conf disabled. Use the manually get h current caret po toggle autom

STEP 4: Revoke SELECT Permission && Revoke ALL PRIVILEGES

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

MANAGEMENT

- Server Status
- Client Connections
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Object Info Session

Week2 (DDL)* DCL Commands* Weeks2(TCL)

```

10 ON bookflow_db.bank_transactions
11 TO 'BranchManager'@'localhost';
12 SELECT CURRENT_USER();
13 REVOKE SELECT
14 ON bookflow_db.bank_transactions
15 FROM 'Auditor1'@'localhost';
16

```

Result Grid

Filter Rows:

Result Grid

Form Editor

Result 1 x

Read Only Context Help Snippets

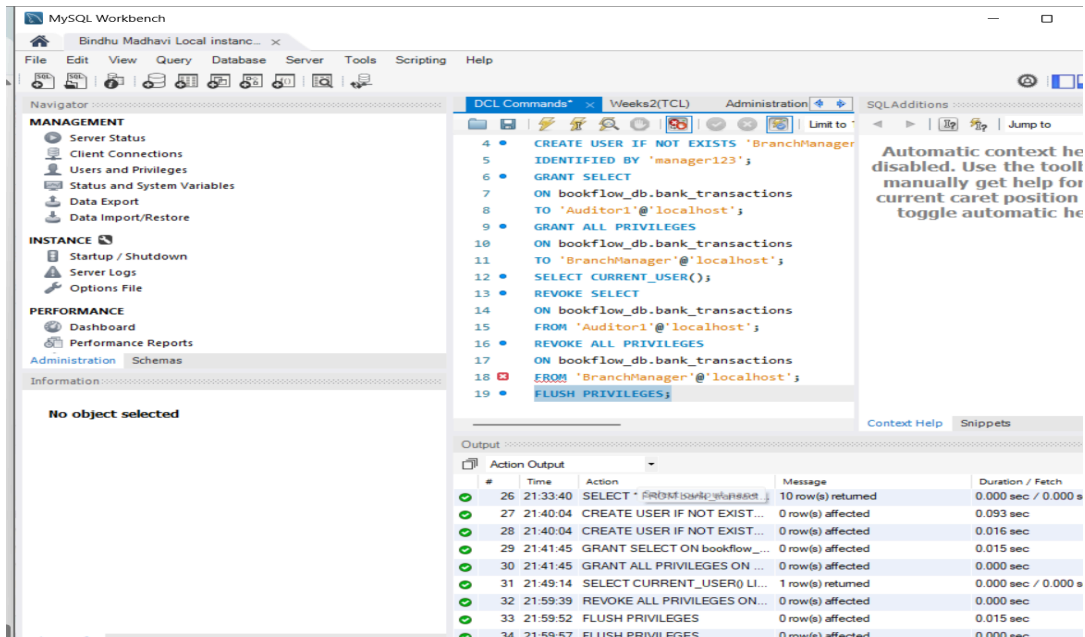
Output

Action Output

#	Time	Action	Message	Duration
23	21:32:53	DELETE FROM bank_transacti...	1 row(s) affected	0.000 se
24	21:33:40	SELECT * FROM bank_transact...	10 row(s) returned	0.000 se
25	21:33:40	SELECT * FROM bank_transact...	6 row(s) returned	0.000 se
26	21:33:40	SELECT * FROM bank_transact...	10 row(s) returned	0.000 se
27	21:40:04	CREATE USER IF NOT EXIST...	0 row(s) affected	0.093 se
28	21:40:04	CREATE USER IF NOT EXIST...	0 row(s) affected	0.016 se
29	21:41:45	GRANT SELECT ON bookflow...	0 row(s) affected	0.015 se
30	21:41:45	GRANT ALL PRIVILEGES ON ...	0 row(s) affected	0.000 se
31	21:49:14	SELECT CURRENT_USER() LI...	1 row(s) returned	0.000 se

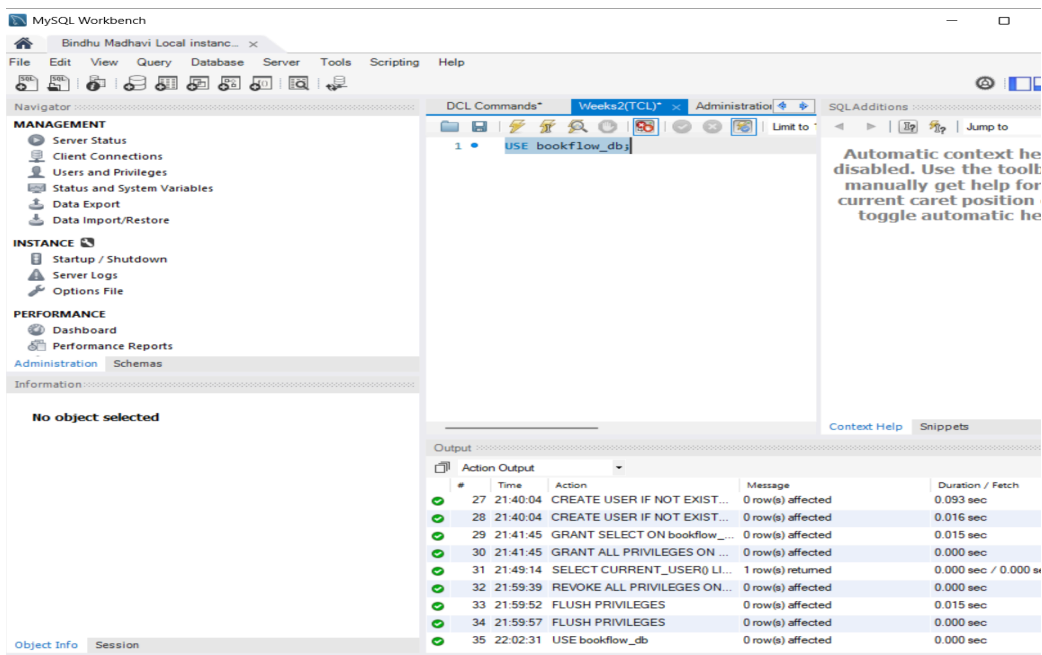
Automatic conf disabled. Use the manually get h current caret po toggle autom

STEP 5: Refresh Privileges

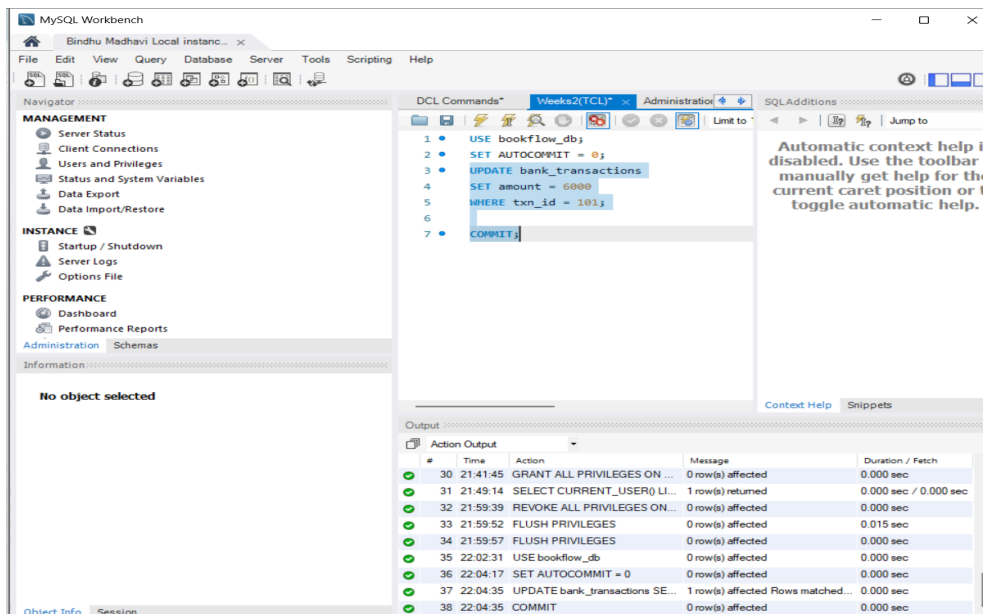


TCL COMMANDS

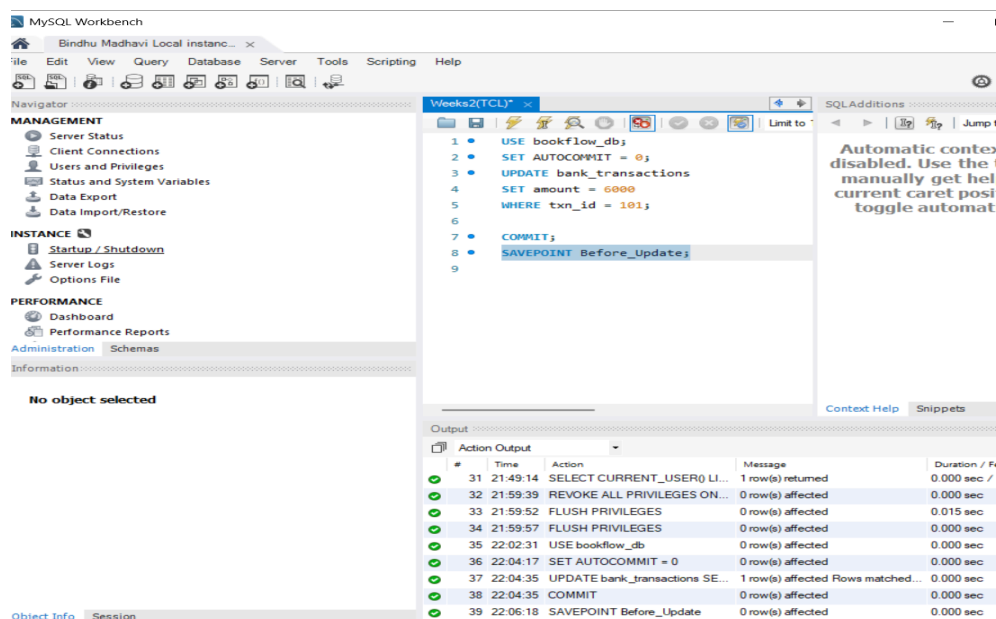
STEP 1: Select the Database



STEP 2: Disable Auto Commit && Update Record



STEP 3: Commit the Transaction & Create a Save Point



STEP 4: Update Another Record

MySQL Workbench interface showing a transaction script in the SQL editor. The script includes a `SAVEPOINT` and an `UPDATE` statement. The output pane shows the execution results.

```

1 • USE bookflow_db;
2 • SET AUTOCOMMIT = 0;
3 • UPDATE bank_transactions
4   SET amount = 6000
5   WHERE txn_id = 101;
6
7 • COMMIT;
8 • SAVEPOINT Before_Update;
9 • UPDATE bank_transactions
10  SET amount = 99999
11  WHERE txn_id = 102;

```

Output:

#	Time	Action	Message	Duration / Fetch
32	21:59:39	REVOKE ALL PRIVILEGES ON...	0 row(s) affected	0.000 sec
33	21:59:52	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
34	21:59:57	FLUSH PRIVILEGES	0 row(s) affected	0.000 sec
35	22:02:31	USE bookflow_db	0 row(s) affected	0.000 sec
36	22:04:17	SET AUTOCOMMIT = 0	0 row(s) affected	0.000 sec
37	22:04:35	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
38	22:04:35	COMMIT	0 row(s) affected	0.000 sec
39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec

STEP 5: Rollback To Save Point

MySQL Workbench interface showing a transaction script in the SQL editor. The script includes a `ROLLBACK TO` statement. The output pane shows the execution results.

```

1 • USE bookflow_db;
2 • SET AUTOCOMMIT = 0;
3 • UPDATE bank_transactions
4   SET amount = 6000
5   WHERE txn_id = 101;
6
7 • COMMIT;
8 • SAVEPOINT Before_Update;
9 • UPDATE bank_transactions
10  SET amount = 99999
11  WHERE txn_id = 102;
12 • ROLLBACK TO Before_Update;

```

Output:

#	Time	Action	Message	Duration / Fetch
33	21:59:52	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
34	21:59:57	FLUSH PRIVILEGES	0 row(s) affected	0.000 sec
35	22:02:31	USE bookflow_db	0 row(s) affected	0.000 sec
36	22:04:17	SET AUTOCOMMIT = 0	0 row(s) affected	0.000 sec
37	22:04:35	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
38	22:04:35	COMMIT	0 row(s) affected	0.000 sec
39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
41	22:09:15	ROLLBACK TO Before_Update	0 row(s) affected	0.000 sec

STEP 6: Full TCL Example

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
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- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Weeks2(TCL)*

```

11 WHERE txn_id = 102;
12 • ROLLBACK TO Before_Update;
13 • UPDATE bank_transactions
14 SET amount = 7000
15 WHERE txn_id = 101;
16
17 • SAVEPOINT SP1;
18
19 • UPDATE bank_transactions
20 SET amount = 9000
21 WHERE txn_id = 102;
22
23 • ROLLBACK TO SP1;
24
25 • COMMIT;
26

```

Automatic context help disabled. Use the toolbar to manually get help for the current caret position or toggle automatic help

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 38	22:04:35	COMMIT	0 row(s) affected	0.000 sec
✓ 39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
✓ 40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
✓ 41	22:09:15	ROLLBACK TO Before_Update	0 row(s) affected	0.000 sec
✓ 42	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.016 sec
✓ 43	22:11:04	SAVEPOINT SP1	0 row(s) affected	0.000 sec
✓ 44	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
✓ 45	22:11:04	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
✓ 46	22:11:04	COMMIT	0 row(s) affected	0.000 sec

Object Info Session

STEP 7: Output

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

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PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Weeks2(TCL)*

```

21 WHERE txn_id = 102;
22
23 • ROLLBACK TO SP1;
24
25 • COMMIT;
26 • SELECT * FROM bank_transactions
27 WHERE txn_id IN (101,102);

```

Automatic context help disabled. Use the toolbar to manually get help for the current caret position or toggle automatic help

Context Help Snippets

Result Grid

txn_id	customer_name	branch_name	trans
101	P.Bindhu Madhavi	Hyderabad	Deposit
102	P.Bindhu Madhavi	Hyderabad	Withdrawal

transactions 1 x Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
✓ 40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
✓ 41	22:09:15	ROLLBACK TO Before_Update	0 row(s) affected	0.000 sec
✓ 42	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.016 sec
✓ 43	22:11:04	SAVEPOINT SP1	0 row(s) affected	0.000 sec
✓ 44	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
✓ 45	22:11:04	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
✓ 46	22:11:04	COMMIT	0 row(s) affected	0.000 sec
✓ 47	22:12:17	SELECT * FROM bank_transact	2 row(s) returned	0.000 sec / 0.00